CSE110: Programming Language I (Lab)
Summer 2025
Lab Exam - 01
Duration: 30 Minutes

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Name:	ID:	Section:

Question 1 [CO1] [10 Points]

Write a Java program that simulates a simple vending machine. The program begins **by taking two integer user inputs: 1) the user's initial balance 2) Which item the customer wants to purchase.**

Each item is selected by entering a number from 1 to 4, where each number corresponds to a specific item with a fixed price. The items available are: **Item 1 (Chips – 50 Taka), Item 2 (Chocolate – 60 Taka), Item 3 (Juice – 80 Taka), and Item 4 (Sandwich – 100 Taka).** For each purchase attempt, the program must check the following conditions:

- If the balance is insufficient to buy it, the program should print **"`Insufficient balance for selected item.`"**
- **If the input is valid and the balance is sufficient**, it should deduct the item price from the balance.
- If the **remaining is an even number** then it will **print a confirmation message with the remaining price.** If the **remaining balance is odd**, the program should **take another user input** by entering another item number. This second purchase will follow the same rules. After this, the program should display the final balance.

Sample Input	Sample Output
Enter balance: 133 Enter item number (1-4): 2 Enter second item number (1-4): 1	Purchase successfully. Remaining balance: 23
Enter balance: 120 Enter item number: 2	Purchase successfully. Remaining balance: 60
Enter balance: 40 Enter item number: 1	Insufficient balance for selected items.

EXPLANATION for sample input 1: The user started with a balance of 133 and purchased item 2(60 Taka), leaving an odd balance of (133-60)= 73tk. Since the remaining balance was odd, customers had to buy another item: Item 1 (50 Taka), resulting in a final remaining balance of (73-53)=23tk and a successful purchase.